

Starting with data

Peter Verhaar



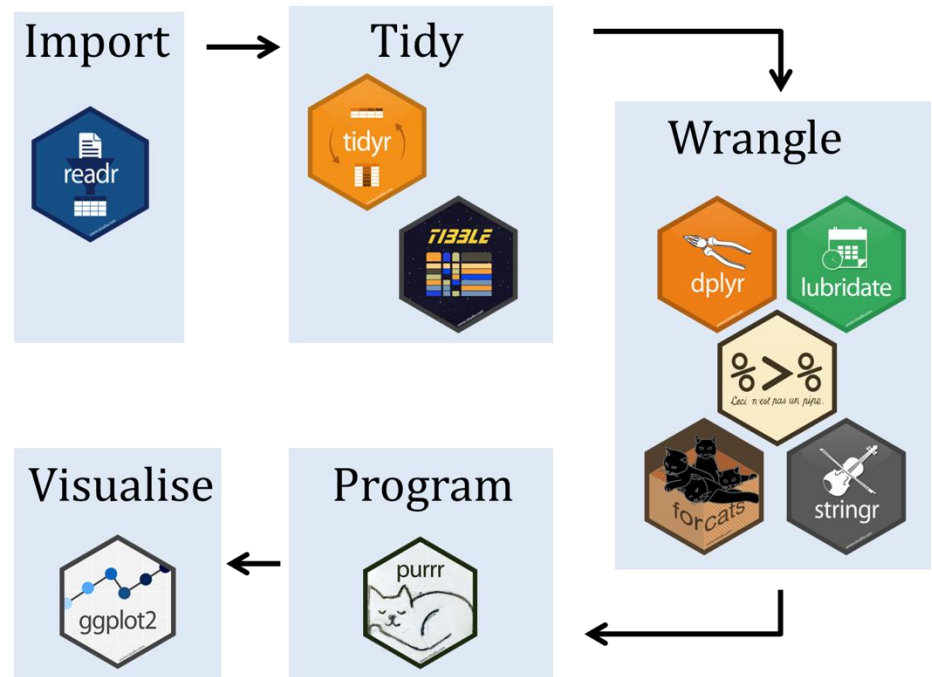
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Recapitulation

- ❑ Objects
(variables):
numbers, strings
- ❑ Naming
conventions for
objects
- ❑ Vectors
- ❑ Functions, e.g.
`mean()` and `max()`
- ❑ Subsetting vectors





“an opinionated collection of R packages designed for data science. All packages share an underlying design philosophy, grammar, and data structures” (<https://www.tidyverse.org/>)

Data frame

- Data structure used to store tabular data
- Each column (variable) is also a vector
- Tidyverse uses the term 'Tibble'

1	"S"	TRUE
7	"A"	FALSE
3	"U"	TRUE
numeric	character	logical

Environment

History

Connections

Tutorial

Import Dataset

58 MiB

List

R

Global Environment

Data

interviews

131 obs. of 14 variables

\$ key_ID	: num [1:131]	1 1 3 4 5 6 7 8 9 ...
\$ village	: chr [1:131]	"God" "God" "God" ...
\$ interview_date	: POSIXct[1:131], format: "2016-...	
\$ no_membrs	: num [1:131]	3 7 10 7 7 3 6 12 ...
\$ years_liv	: num [1:131]	4 9 15 6 40 3 38 7...
\$ respondent_wall_type	: chr [1:131]	"muddaub" "muddaub..."
\$ rooms	: num [1:131]	1 1 1 1 1 1 1 3 1 ...
\$ memb_assoc	: chr [1:131]	NA "yes" NA NA ...
\$ affect_conflicts	: chr [1:131]	NA "once" NA NA ...
\$ liv_count	: num [1:131]	1 3 1 2 4 1 1 2 3 ...
\$ items_owned	: chr [1:131]	"bicycle;televisio..."
\$ no_meals	: num [1:131]	2 2 2 2 2 2 3 2 3 ...
\$ months_lack_food	: chr [1:131]	"Jan" "Jan;Sept;Oc..."
\$ instanceID	: chr [1:131]	"uuid:ec241f2c-060..."

Exercise 1

- Create a new tibble (named ***interview_100***) containing only the data in row 100 of the interviews dataset.
- Create a new tibble (named ***interview_last***) containing only the data from the last row. Use `nrow()` to find the index of the last row. You can use the `tail()` function to check your results.

Exercise 2

- Extract the row that is in the middle of the dataset. Store the data on this middle row in a tibble named ***interview_middle***.
To find the index of the middle row, you can work with the *median()* function, which takes a vector as a parameter. This vector needs to start at index 1, and needs to end at the index of the last row. *Tip*: use the *nrow()* function.
- Combine *nrow()* with the minus ('-') notation to reproduce the behavior of *head(interviews)*, keeping the first 6 rows of the interviews dataset only. Assign the result to ***interview_head***

Exercise 2

- Extract the row that is in the middle of the dataset. Store the data on this middle row in a tibble named ***interview_middle***. To find the index of the middle row, you can work with the *median()* function, which takes a vector as a parameter. This vector needs to start at index 1, and needs to end at the index of the last row.
Tip: use the *nrow()* function.

Exercise 3

- Create code to reproduce the behaviour of `head(interviews)`

Combine `nrow()` with the minus ('-') notation to keeping the first 6 rows of the interviews dataset only. Assign the result to a tibble named **`interview_head`**

Exercise 4

How many respondents are members of an irrigation association?

Make a bar chart, with two values on the X-axis: “No” and “Yes”. The dataset contains the words ‘yes’ and ‘no’ in lower case, but these words need to be capitalised in the plot. Missing values (NA values) can be ignored. Rename the level names using **fct_recode()**

Exercise 5

Add a new column to the interviews data set containing the month in which the interview was conducted. Name this new column 'month'.

Using the `factor()` function, create a bar plot which visualises the number of interviews conducted in the various months. Months in which no interviews were conducted can be ignored.

Data Wrangling

Peter Verhaar

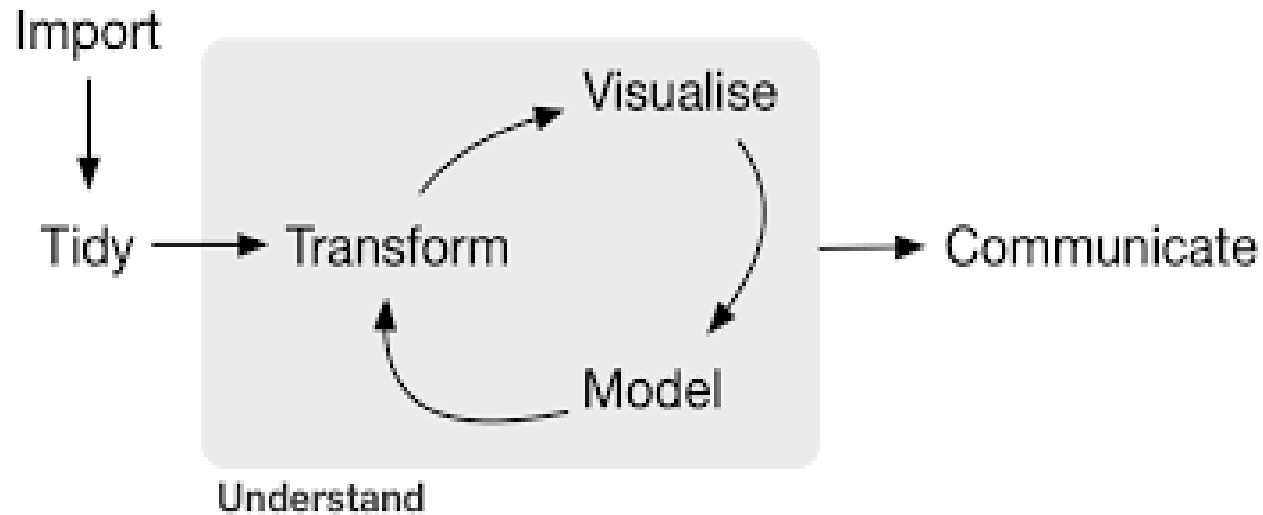


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Data acquisition

Data visualisation



Data analysis

Hadley Wickham, [*R for Data Science*](#) (O'Reilly, 2016)

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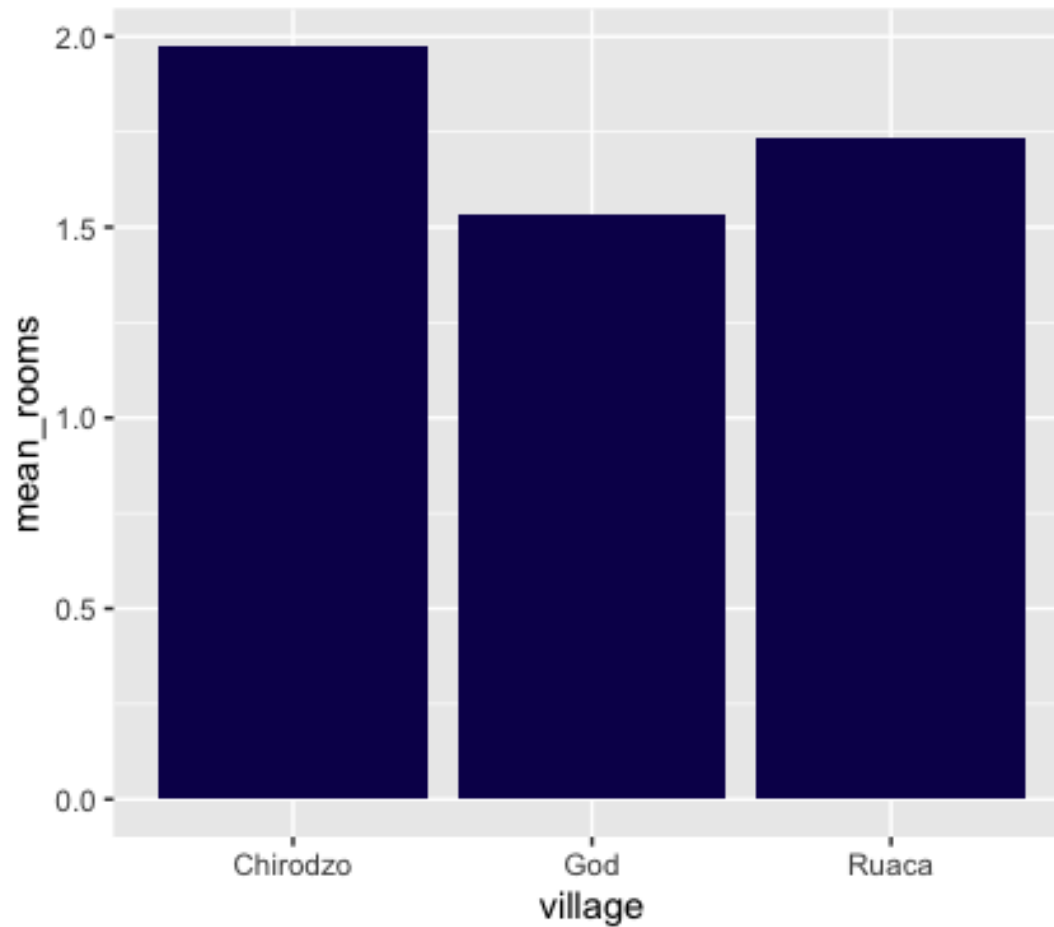
Data

interviews

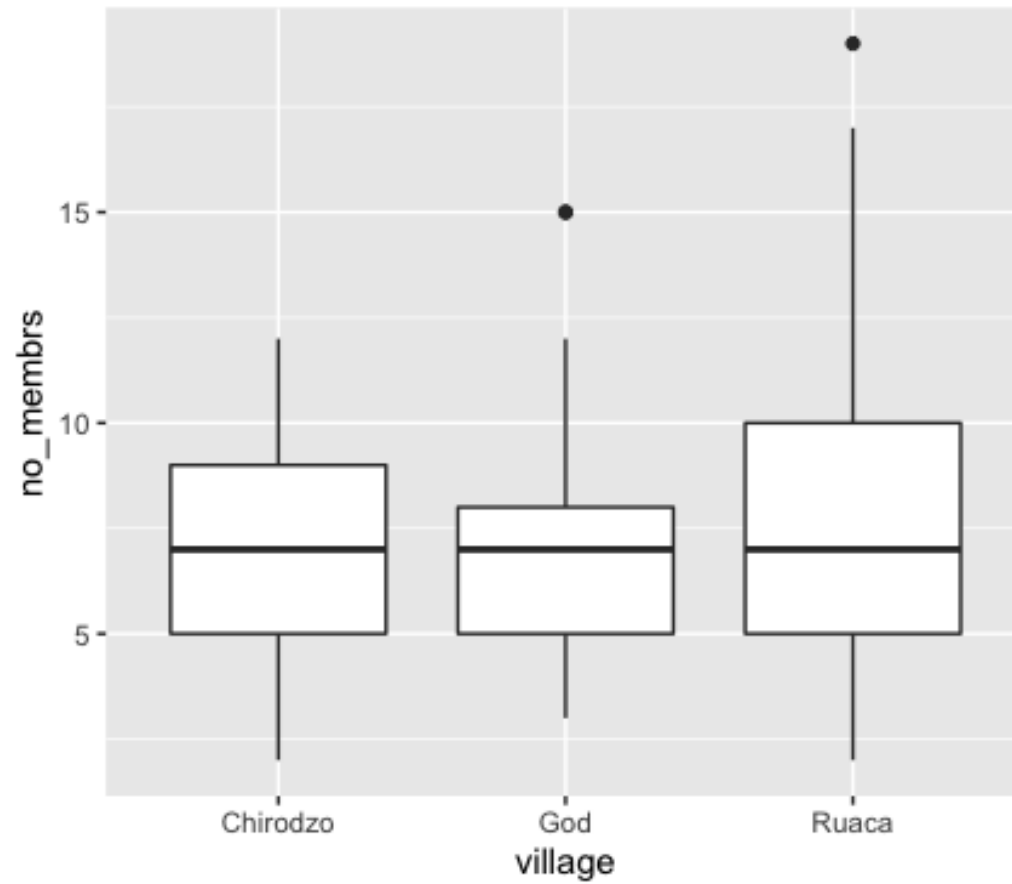
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Average number of rooms in each village



Number of people living in houses





- select
- filter
- mutate
- group_by
- summarise

Exercise 1

Using pipes, subset the interviews data set:

- Include interviews with respondents who were members of an irrigation association only (see the column 'memb_assoc')
- Retain only the columns 'affect_conflicts', 'liv_count', and 'no_meals'.

Exercise 2

Create a new dataframe from the interviews data (name it, for instance 'exercise2') that meets the following criteria:

- Contains only the 'village' column and a new column called 'total_meals'. This new column should specify the total number of meals served in the household per day on average ($\text{no_membrs} \times \text{no_meals}$).
- The final dataframe should only contain the rows where 'total_meals' is greater than 20.
- Hint: think about how the commands should be ordered to produce this data frame!

Exercise 3

Answer the following questions:

- How many households in the survey have two meals per day? How many households have three meals per day? Are there any other numbers of meals represented?

(In other words: create a list of the various options for the number of meals per day, together with a count of the number of households for each of these options)

Tip: Use count()

Exercise 4

Find the mean, min, and max number of household members for each village. Also add the number of observations per village.

(hint: use `group_by()` and `summarize()`. Also use the statistical functions `mean()`, `min()`, `max()` and `n()`).